Decoding Insertions/Deletions via List Recovery

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Department of Computer Science

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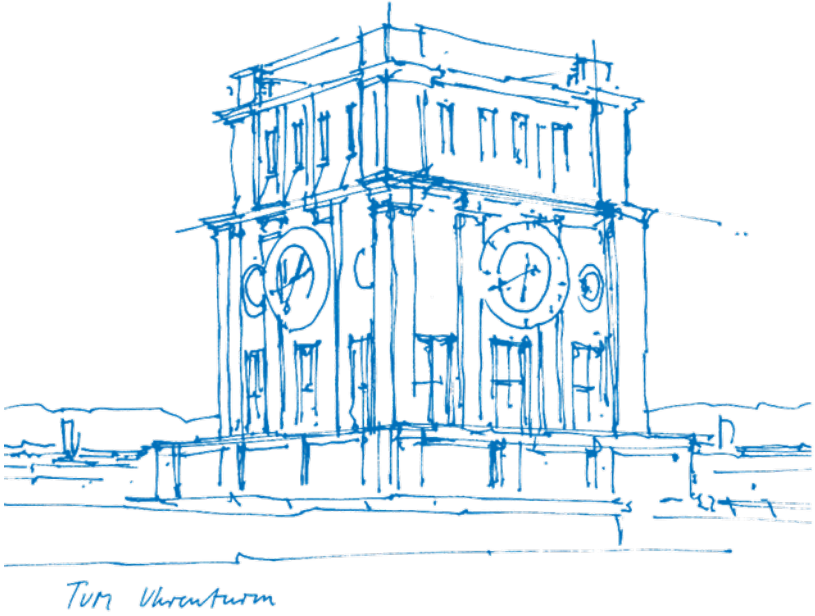

Outline

Introduction

Decoding via List Recovery

- Adversarial Insdels
- Probabilistic Insdels

Adapting Koetter-Vardy Algorithm

Conclusion

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Adversarial Insdels

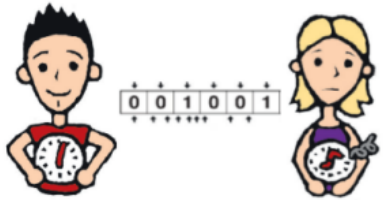
Probabilistic Insdels

Adapting Koetter-Vardy Algorithm

Conclusion

Motivation

- Insertions & deletions occur
  - due to improper synchronization
  - in molecular storage paradigms, e.g. DNA data storage
- Hard to perform decoding efficiently.
- *Trick:* reduce insdel-decoding to an efficiently-solvable problem.
  - E.g., convert to Hamming errors by indexing or reduce to a *list recovery* problem!



G. M. Church *et al.*, "Next-generation digital information storage in dna," *Science*, vol. 337, no. 6102, pp. 1628–1628, Sep. 2012
N. Goldman *et al.*, "Towards practical, high-capacity, low-maintenance information storage in synthesized dna," *Nature*, vol. 494, no. 7435, pp. 77–80, Feb. 2013
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What is List Recovery?

- Consider $\mathcal{C} \subseteq \mathbb{F}_q^n$ that corrects t substitutions.
- Assume integers $t' > t$ and $L, \ell > 1$.

V. Guruswami, "Algorithmic results in list decoding," *Foundations and Trends® in Theoretical Computer Science*, vol. 2, no. 2, pp. 107–195, 2006

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Decoder input

$\boldsymbol{y} =$ | 1 | 3 | $\cdots\cdots$ | 0 | $\in \mathbb{F}_{q=4}^n$

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List-Recoverable Codes

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- A synchronization string ***s*** allows indexing every element in an infinite stream with a constant-sized alphabet.

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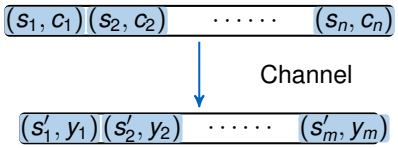

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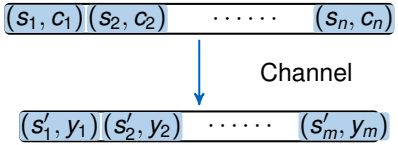

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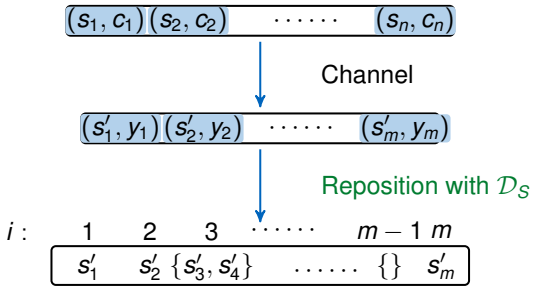

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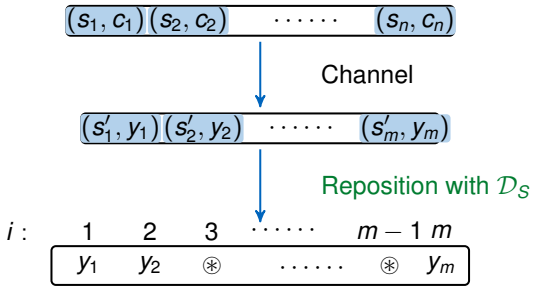

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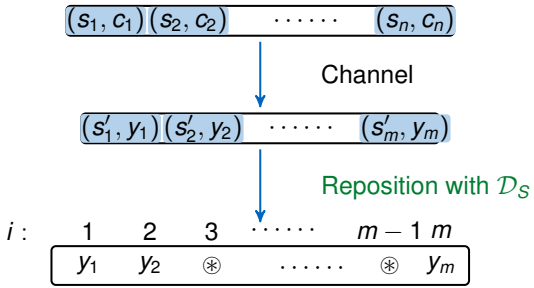

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2. $\mathcal{C}'$ is encodable and decodable in $O(T_{\mathrm{Enc}} + n)$ and $O(T_{\mathrm{Dec}} + n^2(1+\gamma^2)/\epsilon)$ respectively.

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Even More Prior Work

- Reed-Solomon codes for insdels
  - [Safavi-Naini and Wang '02], [Wang *et al.* '04], [Tonien and Safavi-Naini '07], [Duc *et al.* '21], [Liu and Tjuawinata '21], [Con *et al.* '23], [Liu '24], [Con *et al.* '24]

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- Only efficient non-trivial insdel decoder for $[n, k = 2]_q$ RS codes [Singhvi '24]
  - for $[n, k = 2]_q$ RS codes for insdels in [Con *et al.* '24]
  - corrects $n - 3$ deletions in linear time.

Overview of Results

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  - also for jointly decoding multiple received sequences

Introduction

Decoding via List Recovery

Adversarial Insdels

Probabilistic Insdels

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Conclusion

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Consider a $(\rho, \ell = 2, L)$-list-recoverable code $\mathcal{C} \subseteq \mathbb{F}_{q=3}^{n=5}$.

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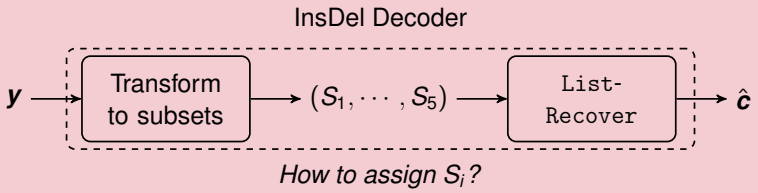

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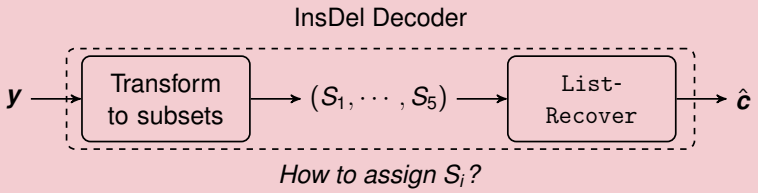

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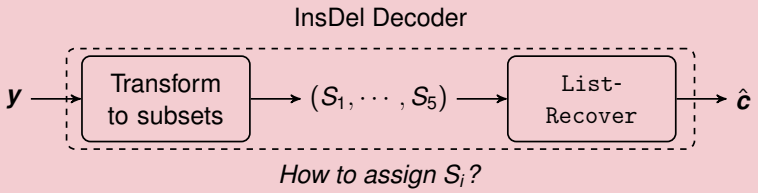

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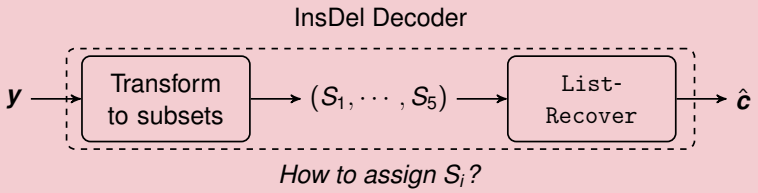

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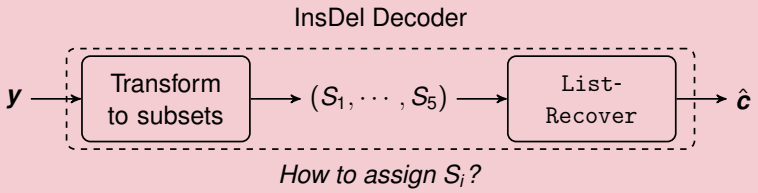

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- Efficiently list-recoverable!

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Probabilistic Insdels

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Conclusion

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Introduction

Decoding via List Recovery

Adversarial Insdels

Probabilistic Insdels

Adapting Koetter-Vardy Algorithm

Conclusion

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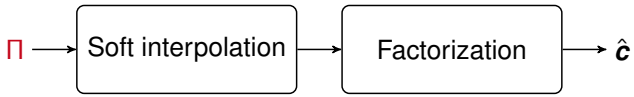

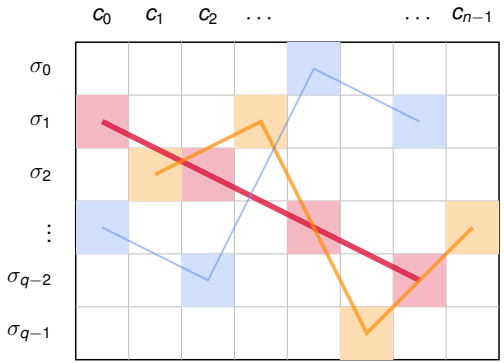

Koetter-Vardy Algorithm

- Soft-decision decoder for RS codes [Koetter and Vardy '03]
  - Recall: $\mathcal{RS}(n,k)_q = \{(f(\alpha_0),\ldots,f(\alpha_{n-1})) \mid f \in \mathbb{F}_q[x], \deg(f) < k\}$,
  - Decoding task: Find best f that fits $(f(\alpha_0),\ldots,f(\alpha_{n-1}))$ best to $\boldsymbol{y}$.
- *Reliability matrix* $\Pi \in (0,1)^{q\times n}$ incorporates soft information on the transmitted symbols.
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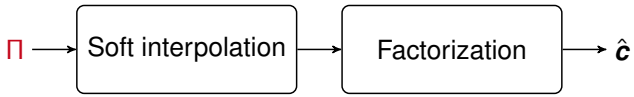

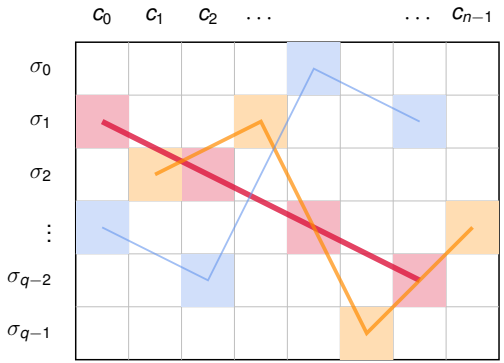

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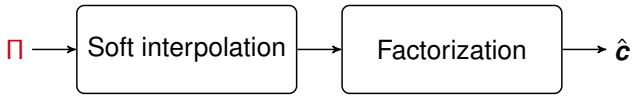

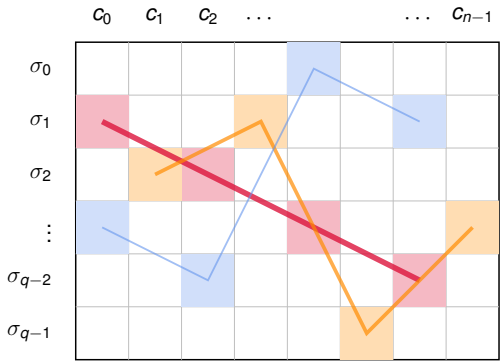

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  - *Joint decoding*: complexity grows linearly in M.

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- Induces random insertions, deletions and substitutions

M. Davey and D. MacKay, "Reliable communication over channels with insertions, deletions, and substitutions," *IEEE Transactions on Information Theory*, vol. 47, no. 2, pp. 687–698, Feb. 2001

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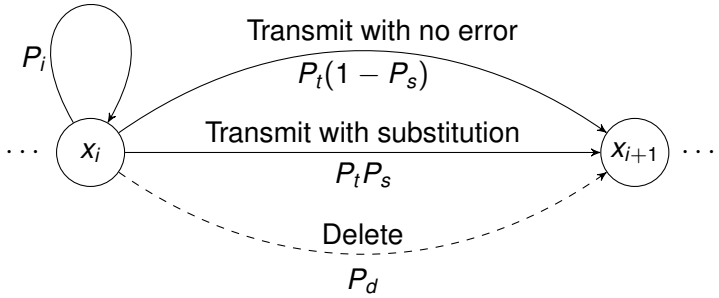

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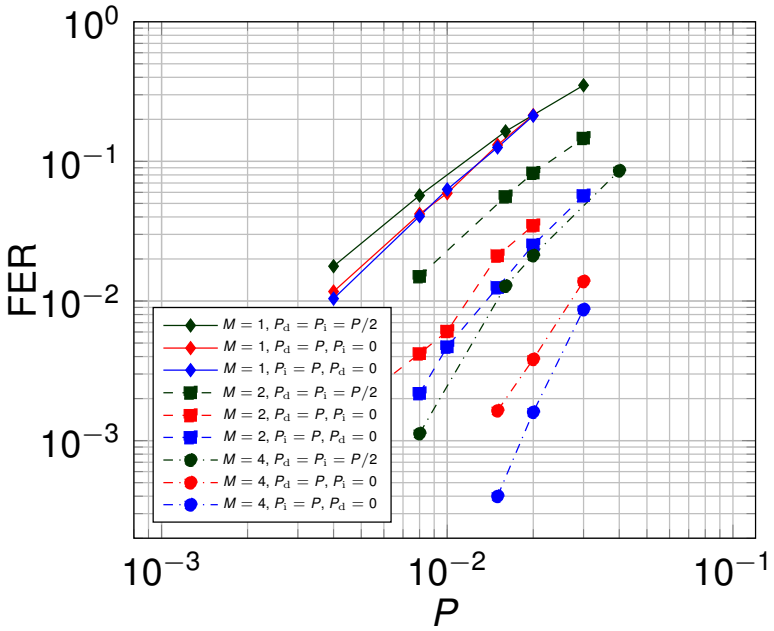

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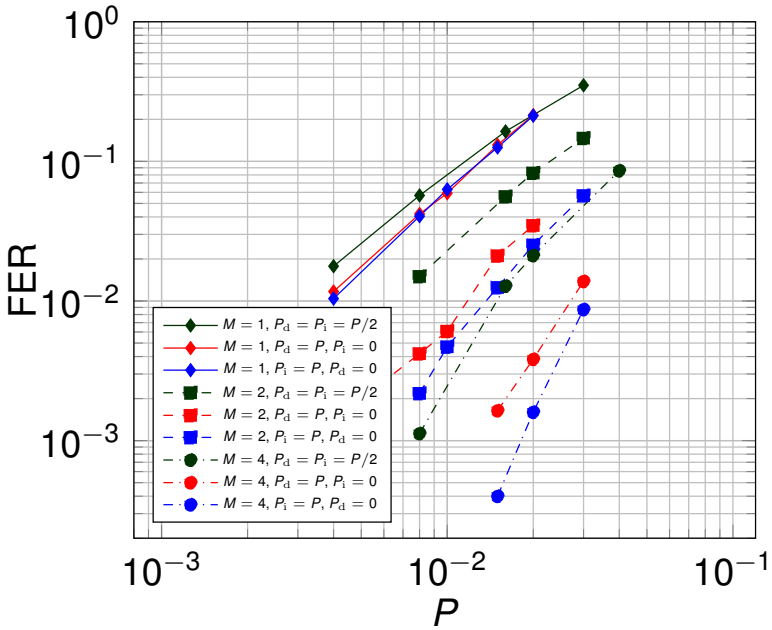

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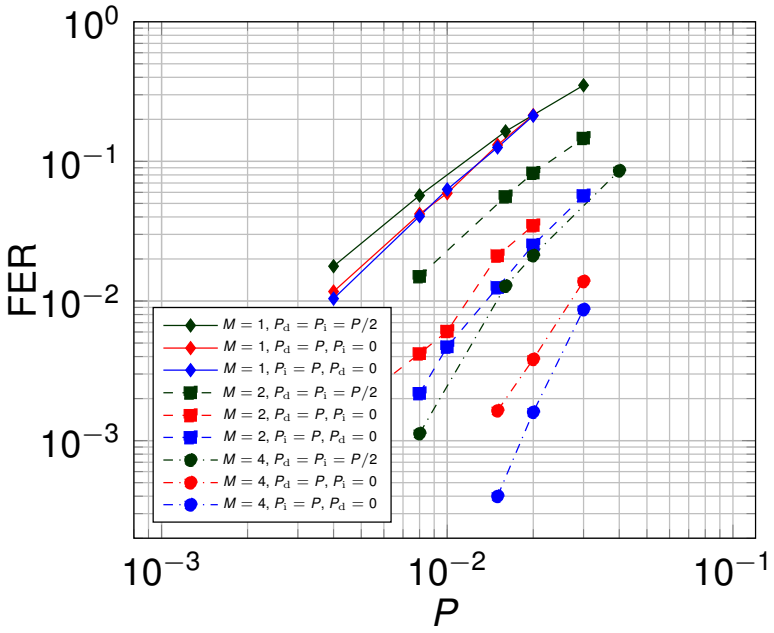
  - As M increases, FER decreases.



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Probabilistic Insdels

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Conclusion

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  - Better rate-error tradeoff for the probabilistic model.
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Thank you!

D!DA DNA DATA APPLICATIONS
European Innovation Council
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